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Smart Home HAR for Speech-Impaired & Hearing-Impaired Users

23CSE399 – Project Phase 1 (Panel Review 2) March 30, 2026

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1 Problem Definition & Relevance

Real-World Use Cases: Resolving Conflicting Interpretations

Because single sensors suffer from context blindness, they critically misclassify transitions between physical states. Our multi-modal fusion architecture specifically resolves these conflicts:

- **UC1: Kitchen (Action → Sign):** A user stops cooking to sign to someone entering the room. An ambient sensor only sees ongoing presence, and a wearable confuses the sign language with cooking arm motions. By gating these inputs with WiFi CSI, the fusion model successfully identifies the shift from a chore to **Communication**.
- **UC2: Coffee (Idle → Sign):** A user sitting idle receives a video call and begins signing. Ambient sensors incorrectly assume a purely static/idle state, while the wearable triggers on sudden, structured arm movement. Fusion combines these to confirm the **Intent to Communicate**.
- **UC3: Resting (Action → Idle):** A user finishes chores and sits down to rest. Wearable macro-acceleration drops to zero, and ambient motion sensors deactivate. The fusion model gates off the WiFi processing to save compute power, safely logging the state as **Idle**.

Motivation:

- **The Voice-First Bias:** Modern smart hubs inherently assume users can speak commands or hear audible alarms, completely excluding speech- and hearing-impaired users from seamless home automation.
- **The Privacy Barrier:** To accommodate sign language, current systems force users to place invasive video cameras in personal spaces, creating an unacceptable trade-off between accessibility and privacy.
- **The Wearable Burden:** While accurate, requiring an elderly or impaired user to wear heavy, full-body sensor suits every day is highly impractical for continuous daily living.
- **Context Blindness:** When non-visual sensors are used in isolation, they suffer from context blindness—easily confusing daily household chores with deliberate communication attempts.

The Ultimate Goal:
Currently, single sensors misclassify **Action** $\rightarrow$ **Sign** transitions as generic continuous movement. A solution is required that intelligently fuses multiple privacy-preserving sensor modalities to isolate deliberate communication intent. By doing so, this project aims to build a silent, invisible home that seamlessly adapts to the user's intent without relying on voice or cameras.

The Need for Fusing Modalities

To achieve these use cases without cameras, we rely on non-intrusive sensors:

- **Ambient Sensors:** Motion detectors that know *where* you are in the house.
- **Minimal Wearables:** Simple sensors that track broad body movements.
- **WiFi Signals (CSI):** Invisible waves that can detect tiny, fine-grained movements like finger gestures.

Using just one of these sensors fails. For example, WiFi can read sign language in the air, but the signal gets completely confused if the person is simultaneously walking around the room. **This is why fusing modalities is our primary focus.** By combining these sensors, we create a system that understands the big picture (macro-movements) and the fine details (micro-gestures) at the same time.

Problem Statement: Activity & Communication

Current smart home systems face a critical trade-off between privacy and functional depth, significantly limiting their real-world applicability. While camera-based architectures offer high recognition accuracy, they severely compromise user privacy in personal spaces; conversely, traditional non-visual sensors preserve privacy but are restricted to detecting only basic, isolated actions. Consequently, these privacy-centric systems fail to navigate the complexity of natural human behavior, particularly when multiple activities occur simultaneously within a household. When users engage in concurrent tasks, existing frameworks lack the multi-modal intelligence required to disentangle and accurately classify these parallel data streams. This limitation poses a significant barrier to inclusivity, as fine-grained, deliberate actions—such as sign language communication—are frequently misclassified or missed entirely amidst everyday background motion.

The problem becomes more difficult in homes with **multiple people**, where current systems cannot reliably determine which activity belongs to which individual. In addition, most smart home systems are reactive in nature. They respond only after an action has occurred and do not have the ability to understand user intent or anticipate future actions. **Overall, there is no existing approach that can reliably handle concurrent activities, identify individuals, and understand user intent while maintaining user privacy in a real home environment.**

Research Objectives

To solve this problem, the project focuses on four clear research objectives:

1. **Multi-Modal Sensor Fusion:** To combine ambient, wearable, and WiFi sensor data into a single intelligent framework capable of tracking complex, parallel activities in a home environment.
2. **Accessible Communication (Sign Language):** To utilize ambient sensor data as a filter to remove ”body movement noise” from WiFi signals, allowing the system to accurately read device-free sign language even when the user is moving.
3. **Predictive Intent Awareness:** To move beyond just recognizing what the user is doing right now. By learning the sequence of a user’s daily habits, the system will predict their future intent (e.g., predicting they are going to bed) to proactively adjust the home environment.
4. **Real-Time Edge Actuation:** To deploy this combined activity-and-communication model onto local hardware, triggering immediate smart home responses (like turning on lights or sending SMS alerts) without relying on cloud processing.

2 Literature & Market Survey

Literature Survey The following table summarizes key datasets used to benchmark HAR, categorized by sensor modality, including datasets focused on fine-grained gestures and predictive intent.

Type	Dataset	Description & Sensors	Critical Limitation
Ambient	**CASAS** (Cook et al.)	The gold standard for smart homes. Uses PIR motion, door, and temp sensors in real apartments.	**Identity Ambiguity:** Cannot distinguish *who* triggered a sensor in multi-resident homes.
Ambient	**ARAS** (Alemdar et al.)	Collected in 2 real houses with multiple residents using 20 ambient sensors.	**Sensor Noise:** High false positives from pets/visitors; lacks ground truth for overlapping activities.
Wearable	**UCI-HAR** (Anguita et al.)	Smartphone (Waist) accelerometer & gyroscope data for 6 basic ADLs.	**Location Blind:** Detects "Walking" but not "Where" (e.g., Kitchen vs. Bedroom).
Wearable	**Opportunity** (Roggen et al.)	72 sensors on body/objects capturing complex gestures (opening doors, drinking).	**Impracticality:** Users must wear a full body suit of sensors; unrealistic for daily living.
Wearable	**UniMiB-SHAR** (Micucci et al.)	Smartphone accelerometer data focused on fall detection and basic activities.	**Limited Granularity:** Good for falls, but poor at distinguishing fine-grained ADLs (e.g., eating).
WiFi	**Widar 3.0** (Zheng et al.)	Large-scale WiFi CSI dataset for gesture recognition using commodity routers.	**Single User Focus:** Accuracy drops significantly when multiple users move simultaneously.
WiFi	**CSI-Bench** (Wang et al.)	Robustness benchmark across different domains (rooms/environments).	**Domain Shift:** Models trained on Room A fail in Room B without extensive retraining.
WiFi	**WIMANS** (Gu et al.)	Multi-user dataset using dual-band WiFi CSI and synchronized cameras.	**Computational Load:** Requires heavy processing to separate mixed signals from multiple bodies.
WiFi	**SignFi** (Ma et al.)	Fine-grained finger gesture recognition (Sign Language) using WiFi CSI.	**Short Range:** Only effective at very close range ($<$ 1m); not suitable for whole-home tracking.
WiFi	**TED-Net** (Wang et al.)	Focuses on privacy-preserving Fall Detection using WiFi skeleton formation.	**Hardware Specific:** Relies on specific router configurations that are hard to deploy universally.

Type	Dataset	Description & Sensors	Critical Limitation
WiFi	**WiFinger** (Li et al.)	Fine-grained finger and gesture recognition using CSI, serving as a foundation for SignFi-like systems.	**Stationary Assumption:** Fails catastrophically when micro-gestures are masked by concurrent macro-movements (e.g., walking).
Ambient	**Aruba (CASAS)** (Cook et al.)	Longitudinal ambient dataset spanning several months, utilized for sequence modeling and intent prediction.	**Concept Drift:** Models predicting user intent fail as daily habits naturally change over time without continuous edge-learning.

Market Survey (Competitor Analysis): We analyzed existing commercial solutions to identify gaps in current technology.

Competitor	Primary Focus	The Research Gap	Reference Link
Smart Home Hubs (Google Nest, Alexa)	**Voice Command & Automation:** Excellent at "if this, then that" triggers (e.g., motion → lights on).	**Context Unaware:** They detect motion but do not understand *activity*. They cannot distinguish "Fall" from "Lying Down".	Frontiers in Robotics & AI
Wearable Tech (Fitbit, Apple Watch)	**Biometrics & Fitness:** Tracking heart rate, steps, and GPS outdoor location.	**Indoor Blindness:** Cannot interact with the home environment (e.g., doesn't know if you are at the stove or the sink).	PMC Healthcare Localization
Security Systems (ADT, Ring)	**Perimeter Safety:** Intrusion detection via door sensors and cameras.	**Privacy & Granularity:** Relies on invasive cameras. Non-camera sensors are binary and miss daily habits.	OVIC IoT Privacy Report
Elderly Monitors (LifeAlert)	**Emergency Response:** Reactive systems (button press) for falls.	**Reactive Only:** Triggers *after* an accident. Cannot predict varying behavior patterns or prevent incidents.	Arquella Care Insights
WiFi Sensing (Origin Wireless)	**Device-Free Sensing:** Using WiFi waves to detect presence/breathing.	**Identity Confusion:** Cannot robustly track multiple people. Signals mix, making it impossible to say *who* is where.	arXiv (Cornell, Jan 2026)

Table 2: Market Survey highlighting the need for Multi-Modal Fusion.

Research Paper Survey:

Paper (Year)	Method / Technique	Description
HAR in Smart Living (2023)	ML/DL-based HAR using multi-sensor data	Comprehensive review of HAR techniques for smart living applications.
IoT Smart Home Automation (2021)	IoT-based automation using sensors and APIs	Proposes an IoT system for automating and controlling smart home devices.
Activity Predictors via Sensor Data (2017)	Imitation Learning with Regression models	Develops models to predict future human activities using sensor data.
SignFi: Sign Language via WiFi (2018)	CNN (9-layer) using WiFi CSI signals	Uses deep learning on WiFi signals to recognize sign language gestures.

Table 3: State of the Art & Open Problems

Takeaway: Existing research treats IoT, predictive modeling, and WiFi sensing as isolated domains. We bridge this gap by unifying them into a single, multi-modal intent-aware framework.

3 Proposed System Architecture

Core Solution: We propose a system integrating **Bazel Board**, **WiFi CSI** (for device-free activity recognition), and **Minimal Wearables** (for identity tagging).

1. **Hybrid Deep Learning Model:** Implement a dual-branch neural network using CNNs to extract spatial features and LSTMs to capture temporal dependencies from the heterogeneous sensor data.

2. **Attention-Enhanced Fusion:** Specific Attention Mechanism within the fusion module to dynamically weight the importance of different modalities (e.g., prioritizing wearables over WiFi for fine motor tasks) before final classification.

3. **Synchronized Preprocessing Pipeline:** Establish a rigid preprocessing workflow focused on Temporal Alignment and sliding-window segmentation to standardize the varying sampling rates between ambient, wearable, and WiFi signals.

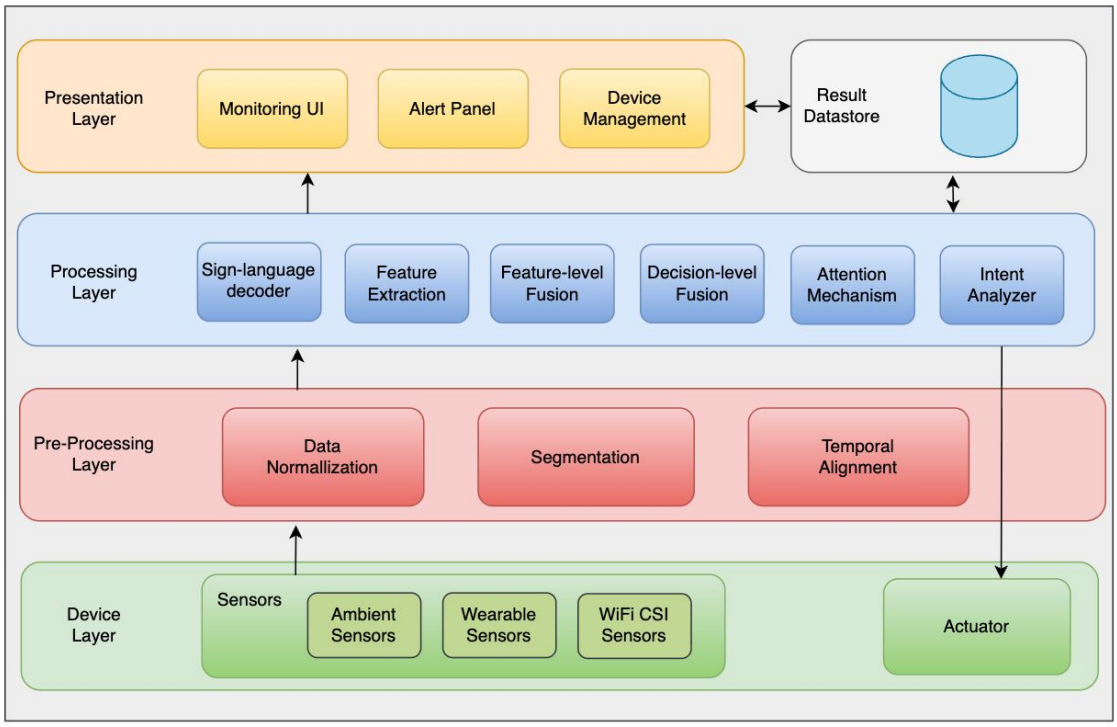


Figure 1: Proposed Multi-Modal Architecture

4 Current Status & Plan

Completed Work (Phase 1 Status):

- **Dataset Preprocessing:** Acquired and preprocessed multiple CASAS smart home datasets (Milan, Cairo, Kyoto2, Kyoto3, Kyoto4). Converted raw sensor logs into padded temporal sequences using activity-based segmentation and numerical encoding suitable for sequential deep learning models.
- **Sequential LSTM Architecture Implementation:** Implemented and systematically compared Uni-LSTM (baseline), Bi-LSTM (bidirectional context), Casc-LSTM (stacked refinement), Ens2-LSTM (ensemble), and CascEns-LSTM (cascade-ensemble) for temporal modeling of ambient smart home data.
- **Model Optimization & Comparative Benchmarking:** Performed hyperparameter tuning (32, 64, 128 units), with 64 units yielding optimal generalization. Cross-validation across selected CASAS datasets showed incremental improvements from bidirectional and ensemble models, while deeper architectures resulted in diminishing returns relative to computational complexity.

Completed Work (Phase 2 Status):

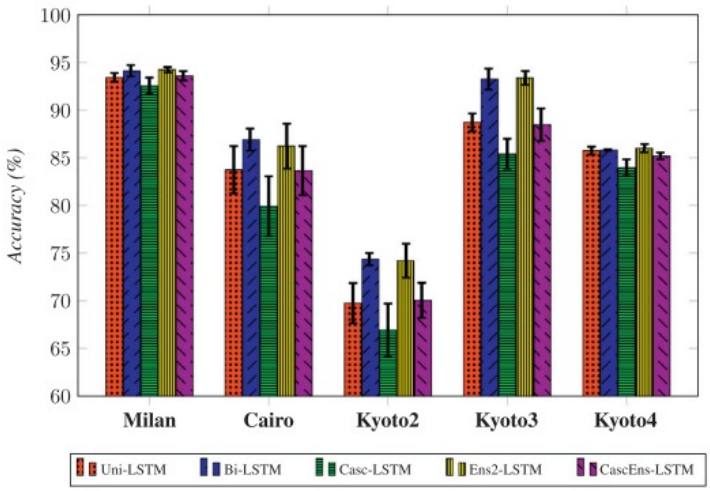


Fig. 9. Averaged accuracy and standard deviation of each dataset over all three folds.

Figure 2: LSTM Model Performance Comparison across evaluated architectures.

Spatial Feature Extraction via CNN

- **Spatial Feature Extraction:** Trained a Convolutional Neural Network (CNN) to extract high-dimensional spatial features (such as Frequency Patterns, Posture changes, etc.) from complex sensor data streams.
- **Dataset Selection (OPPORTUNITY):** Chosen for its exceptional versatility in tracking daily living. It comprehensively covers both short-duration micro-movements (e.g., Open Drawer, Drink from cup) and extended macro-sequences (e.g., Prepare sandwich, Use dishwasher).
- **Performance Benchmark:** The model achieved a classification accuracy of **99.4**% on the OPPORTUNITY dataset.

5 Future Implementation

Next Phase Objectives & System Expansion

- **Proprietary Data Capture:** Construct a homegrown dataset by capturing real-world ambient, wearable, and WiFi CSI sensor streams to validate the system in a custom environment.
- **Hybrid Fusion Models:** Develop an attention-guided architecture to merge the temporal sequence knowledge (LSTM) with the extracted spatial features (CNN) for robust activity recognition.
- **Sign Language Recognition:** Integrate WiFi and wearable-based gesture recognition to specifically accommodate users with speech or hearing disabilities.

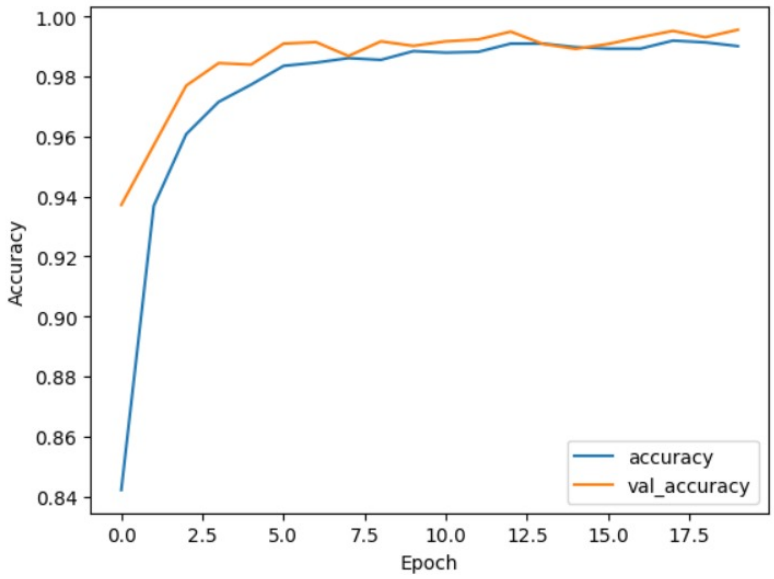


Figure 3: Accuracy graph of CNN on the OPPORTUNITY Dataset.

- **End-User App Development:** Develop a user interface allowing end-users to seamlessly observe system status, interact with the smart home, and receive real-time anomaly alerts.

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